Supplementary Material: Target Generation Model

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1 Target Generation Model

1.1 Software identification and scope

The reference fields used throughout this study are not field measurements but the output of a deterministic propagation generator written for this project, and the contribution evaluated in this paper is therefore the emulation of that generator at graph scale rather than the prediction of measured radio conditions. No commercial or third-party planning tool participates in the pipeline: there is no Atoll, ICS Telecom, WinProp, TIREM or Longley–Rice component anywhere in the chain, and the entire target field is produced by three Python modules and a post-processing corrector that depend, for the propagation arithmetic itself, only on NumPy. Because those modules carry no version string, they are identified here by the SHA-256 digest of their contents, which is the only unambiguous version key available: `generate_realistic_coverage.py` (`95ea0423 3280 6fd6`, 13 631 bytes) holds the propagation equations, `01_data/prepare_transfer_dataset_v19.py` (`45c16d43 bc13 a4bc`, 22 917 bytes) applies them over the terrain graph and writes the five supervised channels, and `data_raw/enrich_rf_targets.py` (`5d38012e c66e 842d`, 26 012 bytes) replaces the two auxiliary channels with terrain-derived attenuation terms. The corrector `contrafactual_alvo_completo.py` (`ebeaf759 2511 fae3`, 10 076 bytes) then rewrote the enriched files as described in Section 1.10; every result of the spatially blocked evaluation uses the corrected files, whereas the comparisons quoted from the earlier campaign use the enriched files before correction. Antenna parameters come from a single ANATEL licensing extract, `antenas_interior_sp_final.csv` (`ad9addb0 c515 a0b4`, 17 763 282 bytes); these records are public ANATEL licensing data, identified in the data package by name, size, and SHA-256 digest and not redistributed there, because they are under a third-party license. Everything reported in this appendix is transcribed from those five files, and the equations below are annotated with the source line at which each expression is implemented so that the correspondence can be checked line by line.

1.2 Base path loss

The base term is a frequency-switched pair of empirical median-path-loss models, both evaluated in decibels with distance in kilometers and frequency in megahertz. The mobile-height correction factor for a medium-to-small city is applied identically in both branches,

$$a(h_m) = \big(1.1\log_{10} f - 0.7\big)h_m - \big(1.56\log_{10} f - 0.8\big), \tag{1}$$

with the mobile height fixed at $h_m = 1.5$ m for every node in the domain (`prepare_transfer_dataset_v19.py:210`). The Okumura–Hata urban reference curve is

$$L_{\mathrm{urb}}^{\mathrm{OH}} = 69.55 + 26.16\log_{10} f - 13.82\log_{10} h_b - a(h_m) + \big(44.9 - 6.55\log_{10} h_b\big)\log_{10} d, \tag{2}$$

and the environment corrections applied to it are

$$L_{\mathrm{base}}^{\mathrm{OH}} = \begin{cases} L_{\mathrm{urb}}^{\mathrm{OH}}, & \text{urban},\\ L_{\mathrm{urb}}^{\mathrm{OH}} - 2\big(\log_{10}\tfrac{f}{28}\big)^2 - 5.4, & \text{suburban},\\ L_{\mathrm{urb}}^{\mathrm{OH}} - 4.78\big(\log_{10} f\big)^2 + 18.33\log_{10} f - 40.94, & \text{rural}. \end{cases} \tag{3}$$

Above 1500 MHz the generator switches to the COST 231 Hata extension,

$$L_{\mathrm{base}}^{\mathrm{C231}} = 46.3 + 33.9\log_{10} f - 13.82\log_{10} h_b - a(h_m) + \big(44.9 - 6.55\log_{10} h_b\big)\log_{10} d + C_m, \tag{4}$$

where $C_m = 3$ dB for the urban class and $C_m = 0$ dB otherwise, and the selection between the two branches is the hard threshold $L_{\mathrm{base}} = L_{\mathrm{base}}^{\mathrm{OH}}$ if $f \le 1500$ MHz and $L_{\mathrm{base}} = L_{\mathrm{base}}^{\mathrm{C231}}$ otherwise (`generate_realistic_coverage.py:85--88`).

Three numerical guards inside these branches change the effective inputs and must be declared because they are not part of either published model. Frequency is clipped to $[150, 2000]$ MHz in the Okumura branch and to $[1500, 2000]$ MHz in the COST 231 branch (`generate_realistic_coverage.py:34,58`), so that every carrier above 2000 MHz enters the base term as if it were at exactly 2000 MHz, while the vegetation term of Eq. (6) continues to receive the true carrier (`generate_realistic_coverage.py:101`) and the two components of the same sum are consequently evaluated at different frequencies; in the four cities this clip is active for 27.4 % of the Lins transmitters, 26.8 % of Campinas, 24.5 % of Sorocaba and 30.1 % of Bauru, which means that for roughly a quarter of the field the base frequency has been silently substituted. Distance is floored at 10^{-3} km, that is at one meter (`generate_realistic_coverage.py:35,59`), three decades below the 1 km lower bound of the Okumura–Hata validity domain. Base-station height enters unclipped, and

5.1 % of the Lins transmitters, 23.3 % of Campinas, 18.9 % of Sorocaba and 6.7 % of Bauru sit below the 30 m floor of that same domain. The generator is therefore an extrapolation of both empirical models over part of its input range, and the reported accuracy figures should be read as fidelity to this particular extrapolation rather than as agreement with the published models inside their stated envelopes.

1.3 Clutter and environment class

What the generator calls the propagation environment is not a clutter parameter in the usual sense, and this is the single point in the model most likely to be misread. There is no land-cover raster, no morphology database, no building or clutter-height layer, and no per-pixel clutter correction anywhere in the chain; the class is assigned once per transmitter by a threshold on its own carrier frequency, urban when $f > 1800$ MHz, suburban when $800 < f \leq 1800$ MHz and rural when $f \leq 800$ MHz (`prepare_transfer_dataset_v19.py:84--86`). The consequence is that the environment label is a deterministic function of frequency and carries no independent geographic information, so that Eq. (3) does not describe the ground under a given node but only the band of the transmitter serving it. A verifiable corollary of this coupling is that the urban class can never reach the Okumura branch and the rural class can never reach COST 231, which the transmitter records confirm exactly: zero transmitters fall in either impossible combination in any of the four cities. Terrain clutter is therefore unmodeled, and a frequency-indexed correction constant stands in its place.

1.4 Vegetation attenuation

Vegetation enters through a single additive term parameterized by NDVI taken raw from column 12 of the terrain feature matrix (`prepare_transfer_dataset_v19.py:140`), which the code maps to a density and then to an effective path depth,

$$\rho_v = \mathrm{clip}\left(\frac{\mathrm{NDVI}+1}{2},\, 0,\, 1\right), \qquad d_v = 0.3\,\rho_v\,d_m, \quad (5)$$

so that at most 30 % of the geometric path is declared to lie inside vegetation regardless of where the vegetation actually is. The attenuation follows the ITU-R P.833 functional form for a single vegetation entry,

$$A_{\mathrm{veg}} = \mathrm{clip}\big(0.25 f^{0.39}\,\min(d_v, 400)^{0.25},\, 0,\, 40\big)\ \text{[dB]}, \quad (6)$$

with frequency in megahertz and depth in meters, a depth cap of 400 m and an attenuation cap of 40 dB. The depth cap is the dominant feature of this term in practice: for a node with NDVI = 0.5 the cap is reached at a geometric distance of about 1778 m, beyond which A_{veg} is constant in distance and depends only on frequency, taking the values 15.87 dB at 900 MHz, 20.80 dB at 1800 MHz and 24.00 dB at 2600 MHz. Because the argument of the quarter-power is a product of distance and NDVI, the term is also strongly distance-driven below the cap, reaching 7.73 dB at 100 m and 13.74 dB at 1 km for the same NDVI at 900 MHz, so that most of its dynamic range is geometry rather than vegetation. The 40 dB clip is unreachable in this dataset: at the highest carrier present in the four cities, 3550 MHz, and with the depth cap and NDVI both saturated, the term attains only 27.10 dB.

1.5 Terrain attenuation

The terrain term is the component of the generator that most needs an explicit and uncomfortable statement, because its name does not describe its content. Formally it is a piecewise-linear function of a dimensionless slope variable,

$$A_{\mathrm{ter}} = \mathrm{clip}\Big(20\,s + 30\,\max(s - 0.1,\, 0),\ 0,\ 30\Big)\ \text{[dB]}, \quad (7)$$

with s the local slope expressed as a rise-over-run ratio. The V19 stage of the pipeline did not read s from the digital elevation model: it drew it independently for each node from a uniform distribution on $[0.02, 0.15]$ by an unseeded call to the NumPy generator (`prepare_transfer_dataset_v19.py:141`, and identically at `generate_realistic_coverage.py:167`), which made Eq. (7) an independent random offset per node between 0.4 dB and 4.5 dB, with mean 1.99 dB and standard deviation 1.15 dB over the support of that distribution. The corrector of Section 1.10 replaces the draw with the node's DEM slope, stored as feature 1 of the released files, and recomputes the total-loss and RSSI channels by exact difference, so that in every reference field used for the blocked evaluation Eq. (7) is a deterministic function of terrain. Recomputing the corrected total-loss channel from the features stored in the released files reproduces it with a mean absolute error of at most 3.02×10^{-4} dB per cell over 150 000 valid-target nodes in five cells.

1.6 Total loss, link budget and serving cell

The three components are summed without interaction terms,

$$\begin{aligned} L_{\mathrm{tot}}(i,j) = {} & L_{\mathrm{base}}(d_{ij}, f_j, h_{b,j}, e_j) \\ & + A_{\mathrm{veg}}(d_{ij}, \mathrm{NDVI}_i) + A_{\mathrm{ter}}(s_i), \end{aligned} \quad (8)$$

and converted to a received level by a two-term link budget that contains no cable, connector, body or feeder loss and no receive antenna gain,

$$\mathrm{RSSI}(i,j) = P_j + G_j - L_{\mathrm{tot}}(i,j) \quad \text{[dBm]}, \quad (9)$$

where $P_j = 10\log_{10}(1000\,P_j^{\mathrm{W}})$ is the transmitter power in dBm (`prepare_transfer_dataset_v19.py:81`) and G_j is the catalog antenna gain in dBi, applied as a scalar in every direction. The serving cell is then the strongest received level among the transmitters whose

planar distance to the node is within the 30 km inclusion radius,

$$s(i) = \underset{j\,:\,d_{ij} \le 30\,\mathrm{km}}{\arg\max}\ \mathrm{RSSI}(i,j), \quad L_i = L_{\mathrm{tot}}\big(i, s(i)\big), \quad (10)$$

which is a pure best-server rule with no handover hysteresis, no load balancing, no cell-selection offset and no minimum-received-level qualification. Distances are planar rather than geodesic, computed as $\Delta y = 111000\,\Delta\varphi$ and $\Delta x = 111000\,\Delta\lambda\cos\varphi$ in meters (`prepare_transfer_dataset_v19.py:183--185`); the constant 111000 m per degree understates the WGS-84 mean meridian degree of 111194.93 m by 0.175 %, a bias that is negligible against the decibel scale of the model but is nonetheless a systematic contraction of every distance in the dataset. Nodes with no transmitter inside the radius retain the sentinel initialization of −200 dBm and 300 dB, and any node whose best level falls below −150 dBm is subsequently overwritten with a constant −110 dBm (`prepare_transfer_dataset_v19.py:274`), which introduces a point mass in the RSSI distribution that must be excluded before any distributional claim about the target field is made.

1.7 Coverage definition

Coverage is a hard indicator rather than a probability, defined on the serving-cell received level by a single threshold,

$$c_i = \mathbf{1}\big[\mathrm{RSSI}_i > -85\ \mathrm{dBm}\big], \quad (11)$$

so that $H(c_i \mid \mathrm{RSSI}_i) = 0$ exactly and the fifth supervised channel carries no information beyond the fourth. There is no fading margin, no outage probability, no shadowing standard deviation and no receiver sensitivity or noise-figure calculation entering this definition; the threshold is a fixed design constant of the generator. The learned coverage channel is accordingly a derived classification of the received level rather than an independent coverage probability.

1.8 Antenna configuration

Table 1 separates the antenna quantities the generator actually consumes from those that exist in the licensing extract and are discarded, and Table 2 gives the resulting per-city inventory. The essential point for a reader assessing the physical realism of the target is that the radiation pattern is isotropic: azimuth, mechanical and electrical downtilt, half-power beamwidth, front-to-back ratio and polarization are all present in the source records and none is read by the loader (`prepare_transfer_dataset_v19.py:73--79`). Because sectors of the same site share coordinates, discarding azimuth turns each site into a stack of co-located omnidirectional sources that differ only in frequency, power, gain and height, with a mean of 13.5 and a maximum of 21 such sources per site in Lins and a maximum of 49 in Campinas; the best-server rule of Eq. (10) then reduces each stack to whichever member happens to maximize $P_j + G_j - L_{\mathrm{tot}}$. The loader also retains catalog gains of exactly 0 dBi where the licensing record carries no value, affecting 32 transmitters in Campinas, 4 in Sorocaba and 1 in Bauru, and it does not apply the 12 dBi substitution that the standalone coverage script uses for the same situation (`generate_realistic_coverage.py:191`), so the two entry points to the same physics are not numerically identical.

Table 1: Antenna quantities in the target generator, grouped by whether the model consumes them. Column names in typewriter font are fields of the ANATEL licensing extract. The second group is the reason the radiation pattern is isotropic: those fields are populated in the records and the loader never reads them.

Quantity	Source and unit in the model
Used	
Carrier frequency f	`FreqTxMHz`, MHz
Transmit power P	$10\log_{10}(1000P^{\mathrm{W}})$, dBm
Antenna gain G	`GanhoAntena`, dBi, isotropic
Antenna height h_b	`AlturaAntena`, m AGL
Site coordinates	`Latitude`, `Longitude`, deg
Environment class e	derived from f alone
Mobile height h_m	fixed at 1.5 m for all nodes
Present in the records but never read	
Azimuth	`Azimute`, deg
Downtilt	`AnguloElevacao`, deg
Half-power beamwidth	`AnguloMeiaPotenciaAntena`, deg
Front-to-back ratio	`FrenteCostaAntena`, dB
Polarization	`Polarizacao`
Absent from the model altogether	
Feeder, connector, body loss	—
Receive antenna gain	—
Noise figure, thermal noise	—
Interference, SINR	—
Fading, shadowing margin	—

Table 2: Per-city transmitter inventory after the generator's validity filter ($f > 0$ and $P^{\mathrm{W}} > 0$). Sites are unique coordinate pairs. The last column is the share of transmitters whose carrier exceeds the 2000 MHz internal clip.

City	Tx	Sites	f (MHz)	P (dBm)	h_b (m)	Clip
Lins	175	13	788–2680	41.6–49.0	21–75	27.4 %
Campinas	4252	350	778–3550	33.0–53.0	3–74	26.8 %
Sorocaba	1398	120	788–2680	35.1–49.3	3–76	24.5 %
Bauru	794	75	788–2680	40.0–49.0	3–90	30.1 %

1.9 The two auxiliary channels

The five supervised channels written by the V19 pipeline are total path loss, two placeholder channels, serving-cell RSSI and the coverage indicator; the placeholders are written as zeros at that stage (`prepare_transfer_dataset_v19.py:276--282`) and are then filled by the enrichment step, which is part of the pipeline for every dataset used in this study. That step computes a vegetation shadow margin following ITU-R P.833 from the Sentinel-2 NDVI of the node and a knife-edge diffraction loss following ITU-R P.526 from the DEM-derived local elevation range and roughness (`data_raw/enrich_rf_targets`

.py:336--396,533--534), using a specific attenuation $\gamma = 0.2\,\mathrm{NDVI}\,(f_{\mathrm{GHz}}/1.8)^{0.4}$ in dB/m over a depth $d_v = d\ \mathrm{clip}(\mathrm{NDVI} - 0.5, 0, 1)$ capped at 40 dB, and a single knife-edge obstacle placed at the path midpoint with an obstruction height $h_{\mathrm{obs}} = \mathrm{clip}(0.5\,\Delta z_{\mathrm{m}} + 0.5\,r_{\mathrm{m}},\ 0,\ 80)$ in meters, also capped at 40 dB, with the local elevation range Δz_{m} and roughness r_{m} in meters after correction. These two channels are therefore the point at which the digital elevation model and the vegetation index enter the reference field as physically labeled attenuation terms, and they are the targets against which the decoder's vegetation and terrain heads are trained and scored. Two properties of the enrichment must be stated because they differ from the geometry of the total-loss channel: the distance it uses is the distance to the nearest transmitter rather than to the serving cell of Eq. (10) (`enrich_rf_targets.py:546`), and the frequency is a single global 1800 MHz (`FREQ_MHZ_DEFAULT`, line 38) rather than the per-transmitter carrier. The auxiliary channels are consequently consistent with each other but evaluated under a different link geometry and frequency from the total-loss channel, and the manuscript states this explicitly where the three channels are compared.

1.10 Target corrector

Three defects of the enriched files were corrected before the spatially blocked evaluation, all by `contrafactual_alvo_completo.py`, which writes new files and leaves the originals untouched. First, the terrain term of the total-loss channel used the random slope draw described above; the corrector substitutes the node's DEM slope and updates the total-loss and RSSI channels by exact difference, which is possible because Eq. (7) depends only on the node and does not change which transmitter serves it. Second, the enrichment step read the topographic position index column where the diffraction term expects the local elevation range; the corrector reads the local-elevation-range column. Third, the obstruction height added the local elevation range in meters to roughness in normalized units, whose scale differs by city; the corrector converts roughness to meters with the per-city elevation standard deviation. The diffraction channel is then recomputed with the original enrichment function. Sentinel nodes keep their sentinel values in the total-loss and RSSI channels. The 16 corrected reference fields are identified by their SHA-256 digests and can be regenerated with the released generator and corrector.

1.11 What the generator does not model

An emulation claim is only as strong as the honesty of the description of what is being emulated, and several mechanisms that a radio-planning reader would expect are simply absent here. There is no ray tracing and no terrain profile along the link: the base term never sees an elevation sample between transmitter and receiver, so diffraction over real obstacles, Fresnel-zone clearance, line-of-sight testing, multipath and ground reflection have no representation, and the knife-edge term of the enrichment substitutes a single midpoint obstacle whose height is a local-elevation-range proxy rather than a profile maximum. There is no clutter database, as discussed above, and no building layer, so the urban, suburban and rural corrections act as three global constants selected by carrier frequency. There is no stochastic component of any kind in the delivered path loss: the docstring of the propagation routine announces log-normal fading (`generate_realistic_coverage.py:80`) but no random term appears in the body of the function, and the only randomness in the pipeline was the terrain slope draw described above, which the corrector removes from the released fields. There is no noise floor, no receiver noise figure, no interference from neighboring cells and therefore no SINR, so the received level of Eq. (9) is a pure two-term link budget and coverage is a threshold on that budget rather than a link-quality decision. Finally, the geometry is two-dimensional and planar: node and transmitter altitudes never enter, the receiver is fixed at 1.5 m above an unspecified reference surface, Earth curvature is ignored, and the distance uses a constant meters-per-degree factor. Stating these absences precisely is what allows the contribution to be defended, because the quantity the network reproduces is a well-defined and fully specified analytical field, and the accuracy figures in the body of the paper measure emulation fidelity to that field and nothing more.